LAPTOP PRICE PREDICTION USING MACHINE LEARNING

1. Introduction

Laptop Price Prediction is a Machine Learning project that predicts laptop prices based on their specifications. It uses details such as brand, category, processor, graphics, RAM, and storage to estimate the price. The main aim of this project is to understand how Machine Learning algorithms can be used for price prediction.

2. Dataset Description

For this project, I used a dataset named laptops_simple.csv.

- Total Records: 943
- Total Columns: 8
- Target Column: Price
- Missing Values: 0
- Duplicate Records: 0

The dataset contains information about different laptop brands and specifications.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset and identify patterns.
Some important observations:

- Most laptop prices are between ₹40,000 and ₹1,00,000.
- Gaming and Convertible laptops have higher average prices.
- Alienware has the highest average price among the brands.
- RAM and storage have a strong relationship with laptop prices.

4. Data Preprocessing

Before training the models, the dataset was prepared.

- Separated input features (X) and target variable (y).
- Used One-Hot Encoding to convert categorical data into numerical values.
- Split the dataset into 80% training data and 20% testing data.
- Used random_state=42 for consistent results.

5. Machine Learning Algorithms

I used three regression algorithms:

1. **Linear Regression:** Predicts prices by identifying relationships between different features and prices.
2. **Decision Tree Regressor:** Predicts prices using a tree-like structure.
3. **Random Forest Regressor:** Uses multiple decision trees and combines their predictions to improve results.

For Random Forest, I used 200 trees.

6. Model Evaluation and Comparison

I evaluated all three models using MAE, MSE, RMSE, and R^2 Score.

Evaluation Metric	Linear Regression	Decision Tree	Random Forest
MAE	20,286.89	17,990.11	15,278.98
MSE	878,708,366.42	1,173,833,708.81	570,420,761.54
RMSE	29,643.02	34,261.26	23,883.48
R^2 Score	0.7956	0.7270	0.8673

Result: Random Forest Regressor performed better than the other two algorithms. It achieved the lowest MAE and RMSE and the highest R^2 Score of 0.8673.

7. Conclusion

This project helped me understand how Machine Learning can be used to predict laptop prices based on different specifications. I learned about data preprocessing, One-Hot Encoding, train-test splitting, regression algorithms, and model evaluation. After comparing three algorithms, Random Forest Regressor gave the best results with an R^2 Score of 0.8673. Therefore, I selected Random Forest as the final model for laptop price prediction.